

Video Transcript

Module 1: Introduction to the Artificial Intelligence Design Process

Video 2: The First Stage of the AI Design Process (07:04)

We will now review the four stages of the design process, starting with the first one. I will use a certain logic to explain the stages one after the other, to help you understand what is needed to design an AI product. However, each stage is an activity in itself. And all can really be done simultaneously or in a different order than the one I will present them in.

In this first stage, we have to identify a behavior we expect artificial intelligence will do. Basically, the AI without the "artificial", i.e., let's just think about intelligence. The intelligence that we will aim to use eventually in an artificial form. There are two challenging choices that need to be resolved in this first stage. Performance metrics and scope.

I will now introduce these two challenging choices and show some examples on how to address them. I'd say one of the biggest challenges, if not the biggest challenge in the design process is to clearly spell out what are the target intelligence performance metrics that we want to implement with artificial intelligence? And this is difficult because what AI can do is so rapidly evolving. That is, the performance frontier in our chosen domain is most likely evolving rapidly.

An easy illustration of the speed of change of the performance frontier can be seen very clearly in what happened with computer vision in the first decade of the century. You will see on this chart how that recognition rate on a big dataset of images called ImageNet evolved over time. For benchmarking purposes, the human error rate on this classification task is 5% because the images are challenging and difficult. This means that if your choice is to set yourself a performance metric of 5%, then you want to implement an AI that is as good as humans in this task.

And in only five years, as you can see in the chart, from 2010 to 2015, recognition rates went from very bad to superhuman performance. In 2010, the state of the art in artificial intelligence had an error rate of 25%. That is, one in four images in the dataset were incorrectly labeled, far

away from the 5% human error rate. However, in the subsequent five years, AI improved so rapidly, that it surpassed human performance. That is what we call superior human behavior. And imagine what this five-year evolution meant for business models based on computer vision that started in 2010. They had to forecast this evolution and take it into account into their plan. A second and related challenging choice, the second one of this first stage, is to spell out what is the scope of this superior behavior that we are expecting to implement?

For example, in the case of self-driving cars, these classification tasks may give us a hint that it's possible to locate all traffic signals in the road just with computer vision. And, in fact, those performance levels achieved beyond 2015 are consistent with the rapid advancements of self-driving cars and the associated technologies. But there may be unforeseen challenges between existing benchmarks like ImageNet, and what we're trying to achieve in our target domain like self-driving cars.

In fact, there are a number of differences between the ImageNet classification task and what self-driving cars have to do such as the fact that self-driving cars process video sequences and not static images, and therefore have more information, and more information to process. Or the fact that some traffic signals may be hidden. Or the error rate that may be tolerated is definitely not 5%.

So, in this first stage, the more precise you can be for what is important that the artificial intelligence does, the better. But it's okay not to be extremely specific until you've gone through the next stages at least once. Especially if this is the first time, you're thinking about it. In some cases, like in the self-driving case, the models chosen, and the approaches chosen, the selections done may restrict what the self-driving option is implemented. Instead of traffic signals, you may choose to focus on finding pedestrians or bicycles. And then you may decide that to distinguish between them and inanimate objects, it's a good idea to use infrared cameras that help you identify warm objects like people. And in some cases, those warm objects or the infrared signals are a complement that can work together with computer vision, iron out some of the issues, and help you bring down that 5% error rate on the ImageNet case for human.

Therefore, what we need to do in this first stage is to address these two challenging choices. First, the performance metric choice. Understand how the performance frontier may be evolving in our chosen domain, so we can send targets for the AI design. And second, the scope challenge.

Understand what is the scope of what we're asking AI to achieve. What is the intelligence we need in our application? The four stages in the design process may have to be revisited more than once as I hinted before.

So, if this is the beginning of your design process, it's okay to be a bit generic and come back after you've done a first iteration of stages two, three, and four. The second time around you may be better equipped to address these two choices, metrics, and scope. It is also possible that the second time around you pivot your original thinking based on finding from your first iteration, like many self-driving companies did, lowering their standards into finding specific solutions like detecting pedestrians or bicycles sometimes in restricted environments, like in urban environments where many of the accidents occur.

